Module 11

Maintenance and Troubleshooting 🦞

Keeping Your Agent Running Smoothly

🧭 Navigation Chart

By the end of this module, you will be able to:

1. **Follow** a daily and weekly maintenance routine
2. **Monitor and optimize** your token spending
3. **Update** 🦞 OpenClaw safely
4. **Back up and restore** your workspace
5. **Diagnose and fix** the most common errors
6. **Use** the command cheat sheet for quick reference
7. **Know** where to get help from the community

📜 Ship's Logbook

Term	Definition
Token burn	The rate at which your AI conversations consume tokens (and money)
Context compaction	Using `/compact` to summarize conversation history and reduce token usage
Workspace backup	A Git commit of your `~/.openclaw/` directory that captures your agent's current state
Rolling update	Updating 🦞 OpenClaw while the gateway is running, with minimal downtime

Realistic Timelines

Level 1 -- Works Immediately:

Basic conversation, file management, simple research, document summarization

Level 2 -- Requires Hours to Days of Setup:

Email automation, trading bots, social media management, code projects, multi-step workflows

The Reality:

- YouTube demos show polished setups that took **days or weeks** to build
- Budget **2-4 hours/week** for setup and maintenance during your first month
- The investment pays off -- but not on day one
- 5 hours/week saved = **200+ hours/year** ($10,000/year in value)

Daily Maintenance Checklist

This takes about **2 minutes**. Do it every day during your first month.

- ⛵ Gateway is running: `npx openclaw status`
- No errors in logs: `npx openclaw logs`
- Token usage reasonable: check your provider dashboard
- Morning brief was sent: check Telegram
- No pending pairing requests: check dashboard for unknown senders

> *If everything checks out, you are good. If anything is off, see the troubleshooting slides later in this deck.*

Weekly Maintenance Checklist

This takes about **10 minutes**. Do it every weekend.

- Security audit: `npx openclaw security audit --deep`
- Health check: `npx openclaw doctor`
- Fix any issues: `--fix` flags on both commands
- Review API spending: visit your provider dashboard
- 🪸 Workspace backup: `git add -A && git commit -m "Weekly backup"`
- Review core files: are `USER.md` and `MEMORY.md` still accurate?
- Disable unused 🐠 skills: `npx openclaw skills list`
- Check disk space: `df -h`

Cost Reduction Strategies

Strategy	Savings	Effort
Enable prompt caching	~90% off system prompt costs	Low
Use Haiku for heartbeats	$50-190/month	Low
Set heartbeat to 55 min	Stays within cache window	Low
Install ClawRouter	Up to 90% via intelligent routing	Medium
Use `/compact` regularly	~67% per conversation	Low

Cost Reduction Strategies (continued)

Strategy	Savings	Effort
Start new sessions for new topics	Prevents runaway token counts	Low
Disable unused 🐠 skills	Reduces context (3K-14K tokens/call)	Low
Use Sonnet for routine tasks	50-75% cheaper than Opus	Low
Use budget models (Kimi K2.5, MiniMax M2.5)	Up to 95% cheaper than Opus	Low
Set API spending limits	Prevents surprise bills	Low

Setting API Spending Limits

Anthropic:

1. Go to console.anthropic.com -- Settings -- Billing
2. Set monthly usage limit (e.g., $50)
3. Set alert threshold (e.g., alert at $40)

OpenAI:

1. Go to platform.openai.com -- Settings -- Billing
2. Set monthly budget cap

> ***Recommendation:*** *Start with a $50/month limit. Increase only when you understand your usage patterns.*

Updating 🦞 OpenClaw

Release channels: Use **stable** (not beta or dev)

Channel	Use For
stable	Everyone in this course
beta	Early access (expect bugs)
dev	Contributors only

Always back up first:

```
cd ~/.openclaw && git add -A && git commit -m "Pre-update backup"
npx openclaw status --json > ~/pre-update-status.json
```

Updating 🦞 OpenClaw (continued)

Update:

```
npx openclaw update
```

After updating:

```
npx openclaw gateway restart
npx openclaw status
npx openclaw doctor
npx openclaw security audit --deep
```

If an Update Breaks Something

Roll back to the previous version:

```
npm install -g @tinybox/openclaw@[previous-version]
npx openclaw gateway restart
```

> *This is why we back up first. You can also restore your workspace from Git if config files were changed by the update.*

Backup and Restore

Quick backup (Git):

```
cd ~/.openclaw && git add -A && git commit -m "Backup"
```

Full backup (archive):

```
tar -czf ~/openclaw-backup-$(date +%Y%m%d).tar.gz ~/.openclaw/
```

What to Back Up

Directory	Priority	Contains
`~/.openclaw/workspace/`	Critical	Identity, personality, rules, 🪸 memory
`~/.openclaw/config/`	Critical	⛵ Gateway config, 🔑 API keys, channel settings
`~/.openclaw/workspace/skills/`	Important	Custom 🐠 skills you built
`~/.openclaw/sessions/`	Nice to have	Conversation history
`~/.openclaw/logs/`	Nice to have	Activity logs

> *Back up **at least weekly**. Critical directories should be committed to Git after any significant changes.*

Debugging Mindset

Before diving into specific errors, adopt this approach:

Rule 1: Run `npx openclaw doctor --repair` first -- most "my agent is being stupid" issues are actually configuration problems. The doctor command catches and fixes the majority automatically.

Rule 2: One workflow at a time -- get one workflow running end-to-end before adding the next. Every new integration is a new failure mode.

- Set up Telegram? Test it thoroughly before adding Discord.
- Adding a 🐠 skill? Verify it works before installing another.
- Building a cron job? Confirm it triggers correctly before scheduling more.

> ***Master each workflow before stacking.*** *Debugging two broken things at once is exponentially harder than debugging one.*

🚩 Damage Control: Gateway Errors

"Gateway refuses to start" (config schema rejection)

- **Cause:** One invalid key in `openclaw.json` blocks boot entirely
- **Fix:** Run `npx openclaw doctor` -- it identifies the bad key. Fix it, restart.
- **Prevention:** Use `npx openclaw config` commands instead of editing JSON directly

"Gateway not running"

- Check: `npx openclaw status`
- Fix: `npx openclaw gateway start`
- Review: `npx openclaw logs`

"Port already in use"

- Find it with `lsof -i :18789`, then kill or change port: `npx openclaw config set gateway.port 18790`

🚩 Damage Control: Gateway Errors (continued)

"Gateway keeps crashing"

- Check logs for specific errors
- Run: `npx openclaw doctor --repair`
- If a plugin was recently installed, try disabling it (plugins run in-process)
- If persistent: restore config from backup, restart

"Session stuck on busy"

- **Cause:** Session lock not released (compaction/abort failure)
- **Fix:** `npx openclaw gateway restart` (releases all locks)
- **Diagnosis:** `npx openclaw status --json` shows stuck sessions

🚩 Damage Control: API and Channels

"API key invalid"

- Check 🔑 key at your provider's console -- regenerate if needed
- Update: `npx openclaw config set models.providers.anthropic.apiKey "[NEW_KEY]"`

"Rate limit exceeded"

- Wait 1-5 minutes, then retry -- if persistent, reduce heartbeat frequency

"Telegram bot not responding"

- Check ⛵ gateway (`npx openclaw status`), check pairing for pending requests
- Verify token in BotFather

"WhatsApp disconnected"

🚩 Damage Control: Ubuntu System

"systemd not running"

- Run: `sudo systemctl daemon-reload`
- Reinstall service: `npx openclaw gateway install`
- Start: `npx openclaw gateway start`

"System very slow"

- Check disk space: `df -h` (ensure >2 GB free)
- Check memory: `free -h`
- Stop background updates if needed
- Reboot: `sudo reboot`

🚩 Damage Control: Performance

"AI responses very slow (30+ seconds)"

- Test internet connectivity
- Check provider status page
- Use `/compact` to reduce context size
- Switch to a faster model temporarily

"Token counter climbing very fast"

- Use `/compact` to compress conversation
- Use `/new` to start a fresh session
- Disable unused 🐠 skills
- Check if large files are being loaded unnecessarily

Command Cheat Sheet: ⛵ Gateway and Status

```
npx openclaw gateway start        # Start the daemon
npx openclaw gateway stop         # Stop the daemon
npx openclaw gateway restart      # Restart the daemon
npx openclaw tui                  # Open chat interface
npx openclaw status               # Is the gateway running?
npx openclaw doctor               # Health check (add --repair to auto-fix)
npx openclaw logs                 # View logs
```

Command Cheat Sheet: Security Audit

```
npx openclaw security audit --deep        # Full audit
npx openclaw security audit --deep --fix  # Audit and auto-fix
npx openclaw config get gateway.auth.mode          # Check auth settings
npx openclaw config set gateway.auth.token "$(openssl rand -hex 32)"  # Rotate token
npx openclaw config get gateway.bind               # Check bind address
```

Command Cheat Sheet: Sandboxing and Tools

```
npx openclaw config set sandbox.mode "[mode]"      # off / non-main / all
npx openclaw config set sandbox.scope "[scope]"    # session / agent / shared
npx openclaw config set tools.deny "[tool]"        # Deny a tool
npx openclaw config set tools.elevated "off"       # Disable elevated mode
```

Command Cheat Sheet: Configuration

```
npx openclaw config                          # Open configuration (re-runs wizard)
npx openclaw config get gateway.port         # Check port
npx openclaw config set models.providers.anthropic.apiKey "[KEY]"  # Update 🔑 API key
npx openclaw config set agents.defaults.heartbeat.every "[N]m"     # Set heartbeat frequency
npx openclaw config set agents.defaults.heartbeat.model "[model]"  # Set heartbeat model
```

Command Cheat Sheet: Channels and Cron Jobs

```
# Channels
npx openclaw channels add [channel]       # Add messaging channel
npx openclaw channels login               # Re-authenticate
npx openclaw config get channels.[ch].dmPolicy  # Check DM mode

# Cron Jobs
npx openclaw cron list                    # List scheduled jobs
npx openclaw cron add [name] --schedule [cron] --task [desc]
npx openclaw cron remove [name]           # Remove a job
npx openclaw cron disable [name]          # Disable a job
```

Command Cheat Sheet: 🐟 Skills

```
npx clawhub browse        # Browse ClawHub
npx clawhub search [q]    # Search skills
npx clawhub install [n]   # Install a skill
npx openclaw skills list           # List installed
npx openclaw config set skills.entries.[name].enabled true   # Enable
npx openclaw config set skills.entries.[name].enabled false  # Disable
```

Command Cheat Sheet: TUI Slash Commands

Command	Action
`/help`	List all slash commands
`/compact`	Compress conversation to save tokens
`/new`	Start a fresh session
`/model [name]`	Switch AI model
`/think [level]`	Set thinking depth (off/low/medium/high/xhigh)
`/status`	Current session info
`/exit`	Leave the TUI

Command Cheat Sheet: Troubleshooting and Nuclear Options

```
OPENCLAW_LOG_LEVEL=debug npx openclaw gateway start  # Debug-level logs
npx openclaw update                # Update OpenClaw
npx openclaw reset --dry-run       # Preview what reset clears
npx openclaw reset --scope config  # Wipe config/creds/sessions
npx openclaw uninstall             # Complete clean removal
```

> `reset` and `uninstall` are **irreversible**. Always `--dry-run` first. Always back up with Git first.

🚩 Rough Waters: Emergency Quick Reference

Step	Action	Command
1. STOP	Kill the agent	`npx openclaw gateway stop`
2. CLOSE	Lock down access	`npx openclaw config set gateway.bind "loopback"`
3. FREEZE	Rotate all tokens/keys	`npx openclaw config set gateway.auth.token "$(openssl rand -hex 32)"`
4. INVESTIGATE	Review logs and sessions	`npx openclaw logs`
5. RESTORE	Fix, audit, restart	`npx openclaw security audit --deep --fix`

Print this table and keep it next to your computer.

Community Resources

Resource	Best For
Official Docs (docs.openclaw.ai)	Configuration, features, API reference
GitHub (github.com/openclaw/openclaw)	Bug reports, feature requests, source code
🦞 OpenClaw Discord	Real-time help, community discussion
🦞 OpenClaw Reddit	Longer-form discussions, tutorials
ClawHub	🐠 Skills marketplace, community contributions

When Asking for Help

Always include these five things in your help request:

1. What you were trying to do
2. The exact error message
3. Your 🦞 OpenClaw version (`npx openclaw --version`)
4. Your OS (Ubuntu 24.04 LTS)
5. What you have already tried

> *Good bug reports get fast answers. Vague ones get ignored.*

🐠 Skills vs. Agents: Know the Difference

Use a Skill when:

- Task is triggered by a specific command
- Doesn't need its own persistent 🪸 memory
- Doesn't run independently -- it's a capability, not an identity

Use a Separate Agent when:

- Needs its own persistent 🪸 memory or identity (e.g., trading bot with different risk parameters)
- Runs independently and proactively
- Communicates through different channels

Default to skills. *Every agent has its own context window and token costs. Multiple agents = multiple costs. Only create a new agent when you genuinely need independence.*

Course Completion Checklist

You should now have:

- **Your agent** -- a fully configured AI assistant running 24/7
- **Security** -- sandboxing, tool policies, authentication, incident response plan
- **Communication** -- Telegram (and optionally Discord/WhatsApp)
- **Automation** -- morning briefs, cron jobs, proactive heartbeats
- 🐠 **Skills** -- from ClawHub and custom-built
- **Knowledge** -- how it all works, how to maintain it, how to troubleshoot it

What to do next:

- Use it daily (🪸 memory builds over time) and update core files monthly
- Try reverse prompting weekly -- join the 🦞 OpenClaw Discord community

💎 Treasure Chest

1. **Maintenance is not optional** -- daily checks (2 min) and weekly audits (10 min) prevent problems
2. **Monitor your spending** -- set limits and check your provider dashboard weekly
3. **Back up religiously** -- Git commit your workspace at least weekly
4. **Update carefully** -- back up before updating, test after updating
5. **Keep the cheat sheet accessible** -- print it, bookmark it, save it to your phone
6. **The community is your resource** -- Discord, GitHub, and Reddit when you are stuck
7. **Security is ongoing** -- run audits weekly, rotate secrets periodically

🌊 Bon Voyage!

Congratulations -- You Have Completed the Course

> *You're not just running a chatbot. You're running infrastructure. Infrastructure for thought. Infrastructure for action. Infrastructure for a new kind of relationship between humans and AI.*

Welcome to the future. You built it safely. Now go use it.